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Improved equations for core loss prediction under asymmetric triangular excitation waveforms based on improved generalized Steinmetz equation

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ABSTRACT

Magnetic components in power electronic devices are frequently subjected to non-sinusoidal excitation. The conventional Steinmetz Equation (SE equation) exhibits significant limitations in predicting core losses under non-sinusoidal excitation conditions. Additionally, other empirical models suffer from low prediction accuracy due to the asymmetry of excitation waveforms or the influence of extreme temperatures. This issue is particularly pronounced under asymmetric triangular excitation waveforms. This study first conducts a comparative evaluation of mainstream empirical models under asymmetric triangular excitation waveforms, revealing that the improved Generalized Steinmetz Equation (IGSE) delivers significantly better prediction performance for such waveforms. However, it is still constrained by three key factors: duty cycle, temperature, and material properties. Consequently, modifications were made to the original IGSE equation. Firstly, the duty cycle of the original IGSE equation was modified based on the waveform composite hypothesis and analysis of errors arising from the superposition of different waveform energies. Subsequently, by displaying the relationship surface between duty cycle, temperature and core losses, the IGSE equation underwent further joint correction for duty cycle and temperature. This ultimately yielded the new improved Generalized Steinmetz equation with joint temperature and duty cycl correction (DT-IGSE equation). This corrected equation not only achieves an intrinsic correction but also compensates for other temperature-dependent variables. To validate the effectiveness and universality of the DT-IGSE equation, experimental data from four typical core materials were selected for verification. Compared with other models, the DT-IGSE equation demonstrated reduced prediction errors for magnetic losses. In summary, the DT-IGSE equation effectively addresses the issue of insufficient prediction accuracy for magnetic core losses under asymmetric triangular excitation waveforms, providing a more precise predictive model to support the loss-optimized design of magnetic components in power electronic devices.

1. Introduction

In the field of modern power electronics, core losses in MAGNETIC COMPONENTS critically affect system stability and efficiency. Consequently, accurately predicting core losses remains a key challenge in optimizing the design of power electronic converters. Triangular and trapezoidal waves are frequently employed as excitation waveforms in high-frequency power switch designs. Due to their fundamental differences from traditional sinusoidal waves, these waveforms pose significant challenges for core loss prediction.

Currently, simpler core loss models are primarily categorized into two types: loss separation models and empirical-calculation models. The loss separation model divides core losses into three components: hysteresis loss, eddy current loss, and residual loss [1]. This model attempts to calculate total losses by separately determining each component's loss. The other type is the empirical calculation model for core loss, which is a method for estimating core loss based on empirical formulas derived from experimental data or theoretical derivations. There are three common empirical calculation prediction models: the SE equation (Steinmetz-Eq. (SE) [2]), the MSE equation (Modified Steinmetz-Equation, MSE [3]), and the IGSE equation (Improved Generalized Steinmetz Eq. [4]). The SE equation is one of the most well-known empirical equations, and numerous core loss prediction models have been developed based on it. However, these models only yield relatively low prediction errors under sinusoidal excitation waveforms.

$$P_{SE} = k \cdot f^{\alpha} \cdot B_m^{\beta} \tag{1}$$

The k, α, and β are coefficients fitted based on experimental data.

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Since magnetic components in power electronic converters often operate under non-sinusoidal excitation waveforms, a modified MSE equation has been proposed based on the SE equation to calculate core losses under arbitrary waveforms [3]. This method does not require additional parameters beyond the original SE equation. The MSE equation assumes that core losses are related to the rate of change of magnetic flux density and posits that core density losses are determined by the weighted average rate of change of magnetic flux density. The calculation method for the equivalent frequency under arbitrary excitation waveforms is given by Eq. (2):

$$f_{eq} = \frac{2}{\Delta B^2 \pi^2} \int_0^T \left(\frac{dB}{dt}\right)^2 dt \tag{2}$$

Substituting the equivalent frequency into eq. (2) allows calculation of core losses under non-sinusoidal excitation waveforms. Coefficients in the equation are then fitted using experimental data to derive Eq. (3):

$$P_{MSE} = \left(k \cdot f_{eq}^{\alpha-1} \cdot B_m^{\beta}\right) \cdot f \tag{3}$$

According to Reference [5], the MSE equation appears to effectively predict excitation waveforms under non-sinusoidal excitation conditions. It employs a combination of effective frequency and repetition frequency to characterize the frequency properties of non-sinusoidal waveforms, while also incorporating core losses caused by magnetic domain motion. However, this model exhibits several shortcomings. Firstly, when considering magnetic flux waveforms resulting from the superposition of two distinct sinusoidal waves, assumimg two sinusoidal wave with frequencies $\omega_0 = 2\Pi f_0$ and $m\omega_0$. The relationship between magnetic flux density and time is then expressed by Eq. (4):

$$B(t) = A[(1-c)\sin\omega_0 t + c\sin m\omega_0 t] \tag{4}$$

However, as the parametere c approaches 1, the equivalent frequency is significantly overestimated while power loss is underestimated. Furthermore, the MSE equation does not specify which frequency of the sinusoidal wave should be chosen as the fundamental wave [6]. Finally, since the core loss per cycle is intrinsically related to the area of the hysteresis loop [7], the MSE equation fails to account for the influence of magnetization rates within a cycle. By performing only static power fitting, it cannot resolve the entire dynamic magnetization process. Consequently, this model provides an ambiguous description of the hysteresis loop's shape characteristics, which directly contributes to its significant prediction errors.

To overcome the insufficient prediction accuracy of SE and MSE, scholars have proposed the IGSE equation based on these models. This equation accounts for the influence of magnetization history and employs the peak difference as the product term in the integral, preventing the magnetic loss during domain switching from being canceled out. However, the IGSE equation still exhibits certain limitations. Firstly, when duty cycles differ, the prediction accuracy of IGSE shows deviations [8,9]. Second, the absence of a temperature term in the equation leads to significant prediction errors under temperature-dependent conditions [10,11]. Building upon this, this paper considers that both duty cycle and temperature affect the domain movement velocity and the shape of the magnetization trajectory, which in turn influence the non-uniform distribution of the instantaneous values of the magnetic flux density change rate. This leads to prediction deviations in the IGSE equation. Therefore, a joint correction equation for duty cycle and temperature, the DT-IGSE equation, is proposed to enhance the prediction accuracy of core losses.

1.1. Analysis based on the waveform composite

When using the original IGSE model to predict core losses with duty cycle as the independent variable, it was found that the model consistently performed best at a 0.5 duty cycle but exhibited significant deviations at extreme duty cycles [8,12]. Therefore, this subsection first proposes a duty cycle correction approach for IGSE based on the waveform composite hypothesis [13].

When plotting B—H curves at different duty cycles using the MagNet online simulation platform [14], it was observed that under identical parameters, the B—H curves corresponding to different duty cycles exhibit significant variations. At a duty cycle of 0.5—where the excitation waveform is symmetrical—the B—H curve assumes a relatively regular shape [15]. This regularity enables many empirical models to effectively describe microscopic processes through macroscopic parameters.

From another macro perspective, the trend of magnetic flux density variation depends on the duty cycle.[4] Simultaneously, frequency—a crucial variable in empirical formulas—is also related to the duty cycle. Fig. 1(a) illustrates an asymmetric triangular excitation waveform, where the rising edge is denoted as A and the falling edge as B. According to the waveform composition hypothesis, this asymmetric triangular wave can be decomposed into the superposition of two symmetric triangular wave energies.

Assuming $B(t)$ has a period T and frequency f, with the rising edge occupying a proportion D of the entire cycle as the duty cycle, we first decompose the rising and falling edges. Then, by reflecting $B(t)$ about its axis of symmetry, two excitation waveforms with different frequencies are obtained. The frequency f_A of the waveform formed by the decomposed rising edge can be analyzed using Eq. (5):

$$f_A = \frac{1}{2D \cdot T} = \frac{f}{2D} \tag{5}$$

The frequency f_B of the waveform composed of the falling segment is given by Eq. (6):

$$f_B = \frac{1}{2(1-D) \cdot T} = \frac{f}{2(1-D)} \tag{6}$$

From Eqs. (5) and (6), it can be seen that the magnetic loss of an asymmetric triangular excitation waveform at frequency f can be decomposed into the superposition of two symmetric triangular excitation waveforms at different frequencies. This can effectively be regarded as equivalent frequency correction, which also explains the relationship between duty cycle and frequency correction. Relevant calculation

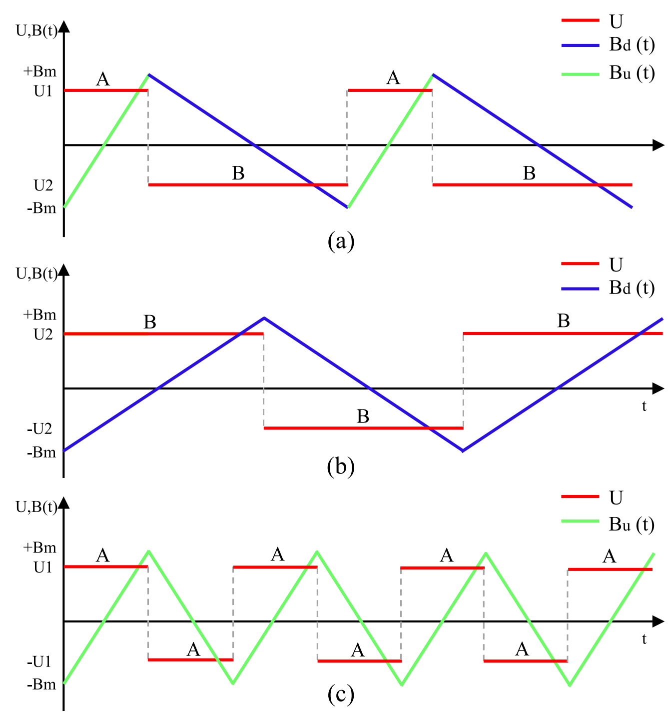

Fig. 1. Process of asymmetric magnetic flux density decomposition.

examples and explanations are also provided in Reference [12].

1.2. Analysis of the combined temperature duty cycle correction principle and process derivation

In the analysis of the waveform composite hypothesis in Section 2, the duty cycle corrected IGSE equation was first derived. The original IGSE equation is given by Eq. (7):

$$P_{IGSE} = \frac{k_i(\Delta B)^{\beta-\alpha}}{T}\int_0^T \left|\frac{dB}{dt}\right|^\alpha dt \tag{7}$$

Where ΔB is the peak-to-peak value of magnetic flux density within one cycle (i.e., the difference between the maximum and minimum values within one cycle), and k, α, β are the coefficients of the Steinmetz equation at the equivalent frequency. At this point, k_i is given by Eq. (8):

$$k_i = \frac{k}{(2\pi)^{\alpha-1}\int_0^{2\pi}|\cos\theta|^\alpha 2^{\beta-\alpha}d\theta} \tag{8}$$

For core losses under a fundamental frequency asymmetric excitation waveform, the formula is given by Eq. (9):

$$P = P_A + P_B \tag{9}$$

Since the relevant frequency derivation and analysis of the decomposed symmetric waveforms were already performed in Section 2, it follows that P_A and P_B are given by Eq. (10):

$$\begin{cases} P_A = f_A\displaystyle\int_0^{\frac{1}{2f_A}} P_{IGSEa}(t)dt \\ P_B = f_B\displaystyle\int_{\frac{1}{2f_B}}^{\frac{1}{f_B}} P_{IGSEb}(t)dt \end{cases} \tag{10}$$

For triangular excitation waveforms, the calculated result of $P_{IGSEa}(t)$ in the first half-cycle is identical to that in the second half-cycle. The core integral term yields the result given by Eq. (11):

$$\int_0^{\frac{T}{2}}\left|\frac{dB}{dt}\right|^\alpha dt = T\left(\frac{\Delta B}{T}\right)^\alpha = \int_{\frac{T}{2}}^T\left|\frac{dB}{dt}\right|^\alpha dt \tag{11}$$

Among these, $P_{IGSEa}(t)$ and $P_{IGSEb}(t)$ represent the IGSE equations corresponding to the magnetic flux density changes in segments A and B after decomposition, respectively. Substituting these into the fundamental frequency period using eq. (11) yields Eq. (12):

$$\begin{cases} P_A = \dfrac{1}{2DT}\displaystyle\int_0^{DT} P_{IGSEa}(t)dt \\ P_B = \dfrac{1}{2(1-D)T}\displaystyle\int_0^{(1-D)T} P_{IGSEb}(t)dt \end{cases} \tag{12}$$

Considering that error effects [13,16] and material coefficient influences may arise when energies of different frequency waves superimpose, it is hereby assumed that an exponential factor should be present in the product term preceding integration. Under this assumption, the magnetic loss equations for Sections A and B are given by Eq. (13):

$$\begin{cases} P_A = \left(\dfrac{1}{2D}\right)^\gamma \dfrac{1}{T}\displaystyle\int_0^{DT} P_{IGSEa}(t)dt \\ P_B = \left(\dfrac{1}{2(1-D)}\right)^\gamma \dfrac{1}{T}\displaystyle\int_0^{(1-D)T} P_{IGSEb}(t)dt \end{cases} \tag{13}$$

After incorporating the index term, one of the preceding product terms in A or B is approximated as zero to simplify calculations when operating at extreme duty cycles. Finally, combining these yields the magnetic loss calculation formula for the fundamental frequency asymmetric waveform as Eq. (14):

$$P = \frac{k}{D^\gamma}\cdot\frac{1}{T}\int_0^T\left|\frac{dB}{dt}\right|^\alpha \Delta B^{\beta-\alpha}dt \tag{14}$$

At this point, the parameter k serves as an influencing factor to assist in modifying the magnetic loss prediction formula. Based on the symmetry relationship of core losses [17], when D is less than 0.5, $D = D$; when D is greater than 0.5, $D = (1 - D)$.

Since temperature variations directly affect the intrinsic properties of magnetic materials, such as magnetic permeability and magnetic flux density, these factors may influence the B—H curve and introduce errors in magnetic loss prediction. Furthermore, these two variables may exhibit certain coupling relationships. Fig. 2 analyzes the duty cycle, temperature, and magnetic core loss surfaces for four materials.

Fig. 2 further validates the aforementioned correction method for the duty cycle factor. Core losses exhibit an approximate exponential relationship with duty cycle variation, while temperature follows a hyperbolic variation pattern at the same duty cycle. Therefore, the temperature factor *TEMP* [18] is incorporated into the duty cycle correction Eq. (15):

$$P_{DT-IGSE} = \frac{k}{D^\gamma}\cdot\frac{1}{T}\int_0^T\left|\frac{dB}{dt}\right|^\alpha \Delta B^{\beta-\alpha}dt + TEMP \tag{15}$$

Where the a and b are the parameters to be fitted, and *temp* is the ambient temperature.

Considering the coupling relationship between material and temperature, we first tentatively introduced the temperature factor as both a multiplicative term and an additive term for fitting and comparison with actual values. It was found that the additive form, combined with Eqs. (14) and (15), proved more suitable. Consequently, the final joint correction equation DT-IGSE is given by Eq. (16):

$$P_{DT-IGSE} = \frac{k}{D^\gamma}\cdot\frac{1}{T}\int_0^T\left|\frac{dB}{dt}\right|^\alpha \Delta B^{\beta-\alpha}dt + TEMP \tag{16}$$

2. Experimental data and results analysis

2.1. Analysis of experimental measurement principles

The measurement of core losses is currently generally performed using the AC power method [19], as shown in Fig. 3. The core under test is typically a toroidal ring (where l_e is the average magnetic path length; A_e is the cross-sectional area of the core). An excitation winding and an induction winding are uniformly wound on the core, with N_1 and N_2 representing the number of turns in the excitation and induction windings, respectively, typically set as $N_1 = N_2 = N$. A signal generator produces a sinusoidal wave or other waveform at a given frequency f (period $T = 1/f$). The net energy stored in the inductor during one cycle is given by Eq. (17):

$$W = \int_0^T u(t)\cdot i(t)dt \tag{17}$$

The high-frequency power amplifier supplies high-frequency excitation to the excitation winding. According to Ampère's circuit law, the excitation current $i(t)$ in the winding generates a magnetic field strength $H(t)$ in the core. This represents the Lorentz force exerted on a unit current element within the magnetic field and serves as a crucial parameter for describing magnetic field intensity. This magnetic field strength acts upon the magnetic core material to produce magnetic flux density $B(t)$, defined as the magnetic flux per unit area perpendicular to the magnetic field lines. According to the electromagnetic induction principle, the alternating magnetic flux in the core induces an induced voltage $u(t)$ in the sensing winding. By measuring the excitation current $i(t)$ of the core winding under test, the magnetic field strength $H(t)$ and magnetic flux density $B(t)$ are obtained. Substituting $H(t)$ and $B(t)$ with $i(t)$ and $u(t)$ and combining with eqs. (17) [20] yields eq. (18):

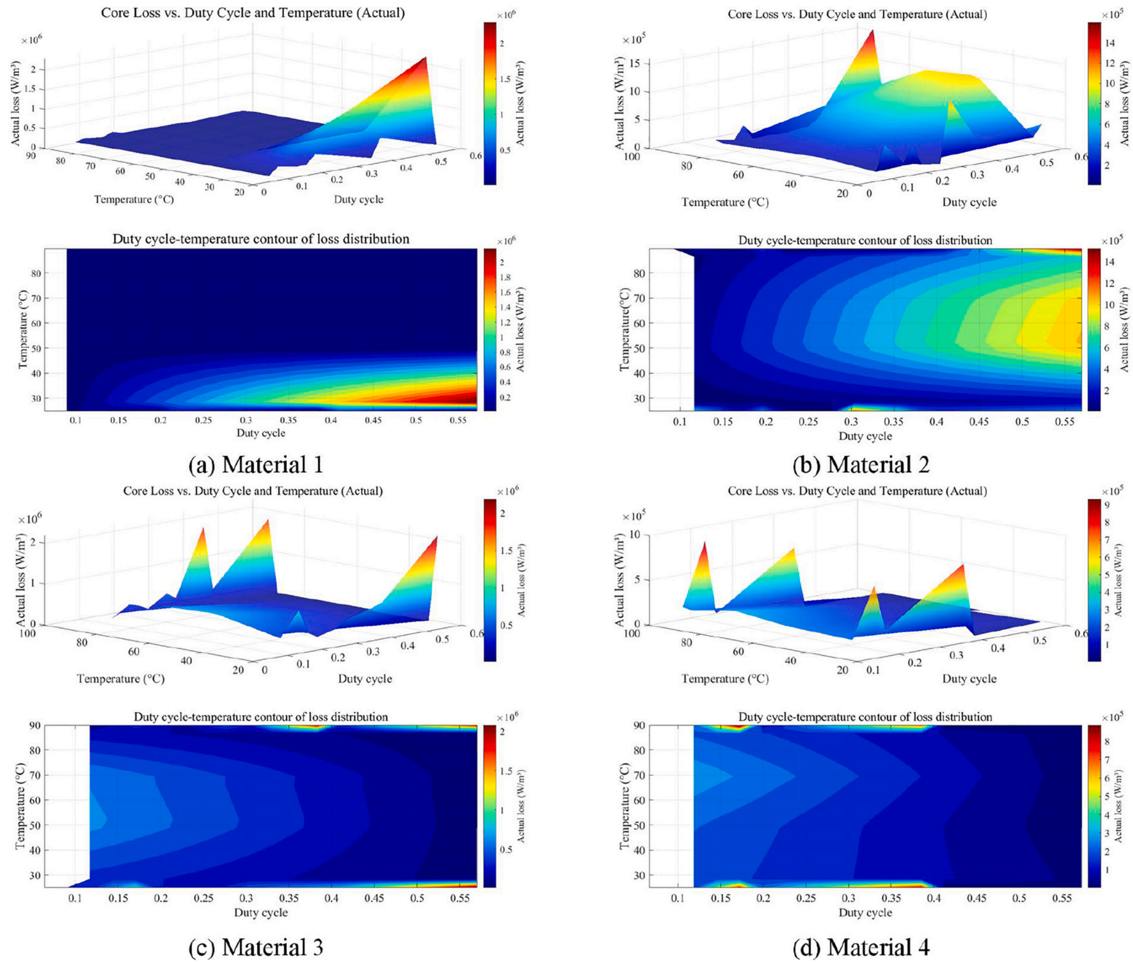

Fig. 2. Surface analysis of core loss vs. duty cycle temperature for different materials.

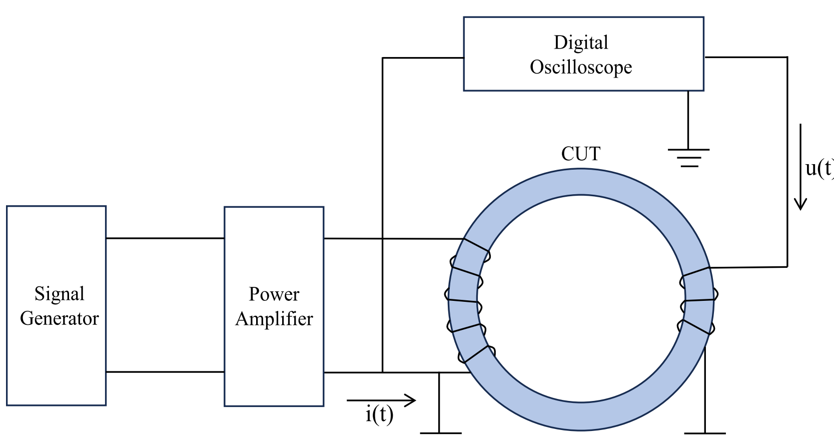

Fig. 3. Double-Winding method for measuring magnetic core loss.

$$W = \int_0^T \left(NA_c\frac{dB(t)}{dt}\right)\left(\frac{H(t)l_m}{N}\right)dt \tag{18}$$

Using eq. (18), the output power per unit volume of the excitation source is calculated [21,22], i.e., the magnetic core loss density P is given by Eq. (19):

$$P = \frac{1}{T}\int_{B(0)}^{B(t)} HdB \tag{19}$$

From eq. (19), it shows that the core loss per unit volume within one excitation cycle equals the area of the B—H hysteresis loop.

Due to the complexity of core materials, they are denoted here simply as Material 1, Material 2, Material 3, and Material 4. Within one cycle, 1024 sampling points representing magnetic flux density are obtained at equal intervals, with approximately 1400 sets of triangular excitation waveform data for each material group.

2.2. Data results analysis

Experimental data collected were compared with MSE, IGSE, and DT-IGSE equation models. Parameter fitting and model comparisons were performed for the four material groups [8], yielding error curves under different duty cycles as shown in Fig. 4.

For parameter fitting, a traversal search is employed to locate the multivariate minimum. Using Eq. (20), The objective value is minimized by finding the minimum error between the function values of the sampling points and the predicted points using Eq. (20):

$$err_{\min} = \min\left(\sqrt{\frac{\sum_{i=0}^{1024}\left(pred(i) - actual(i)\right)^2}{1024}}\right) \tag{20}$$

The corresponding parameters at this point represent the fitting parameters for the current model objective. The method of squaring and then taking the square root prevents positive and negative errors from canceling each other out. Table 1 lists the parameters in the fitted DT-IGSE equation.

To avoid model errors caused by parameter identification, the same fitting method was applied to other three models, with identical initial parameter guesses. For the DT-IGSE equation, the initial guess for the temperature term parameters was set to 0, while γ was initialized to 1. As shown in the error curve distribution in Fig. 4, although the prediction error was slightly higher than the other two models for certain duty cycles in specific materials, the overall error was significantly reduced compared to the other two models. Furthermore, the DT-IGSE demonstrated higher robustness across different materials, making it suitable for a wide range of materials.

Fig. 5 shows the error comparison of different materials using different models at different temperatures. Materials 2, 3, and 4 all exhibit smaller average errors at different temperatures, but Material 1

Table 1
DT-IGSE Fitting Parameters for Different Materials.

Parametric	Materal 1	Materal 2	Materal 3	Materal 4
α	1.8057	1.7169	1.7497	1.8507
β	2.3274	2.3172	2.4380	2.4826
γ	0.0908	0.1377	0.1180	0.163
k	5.302e-4	2.038e-3	1.710e-3	5.198e-4
a	−0.3071	−0.6578	−1.0764	0.8719
b	−0.1721	0.2565	−2.1233	−0.4270

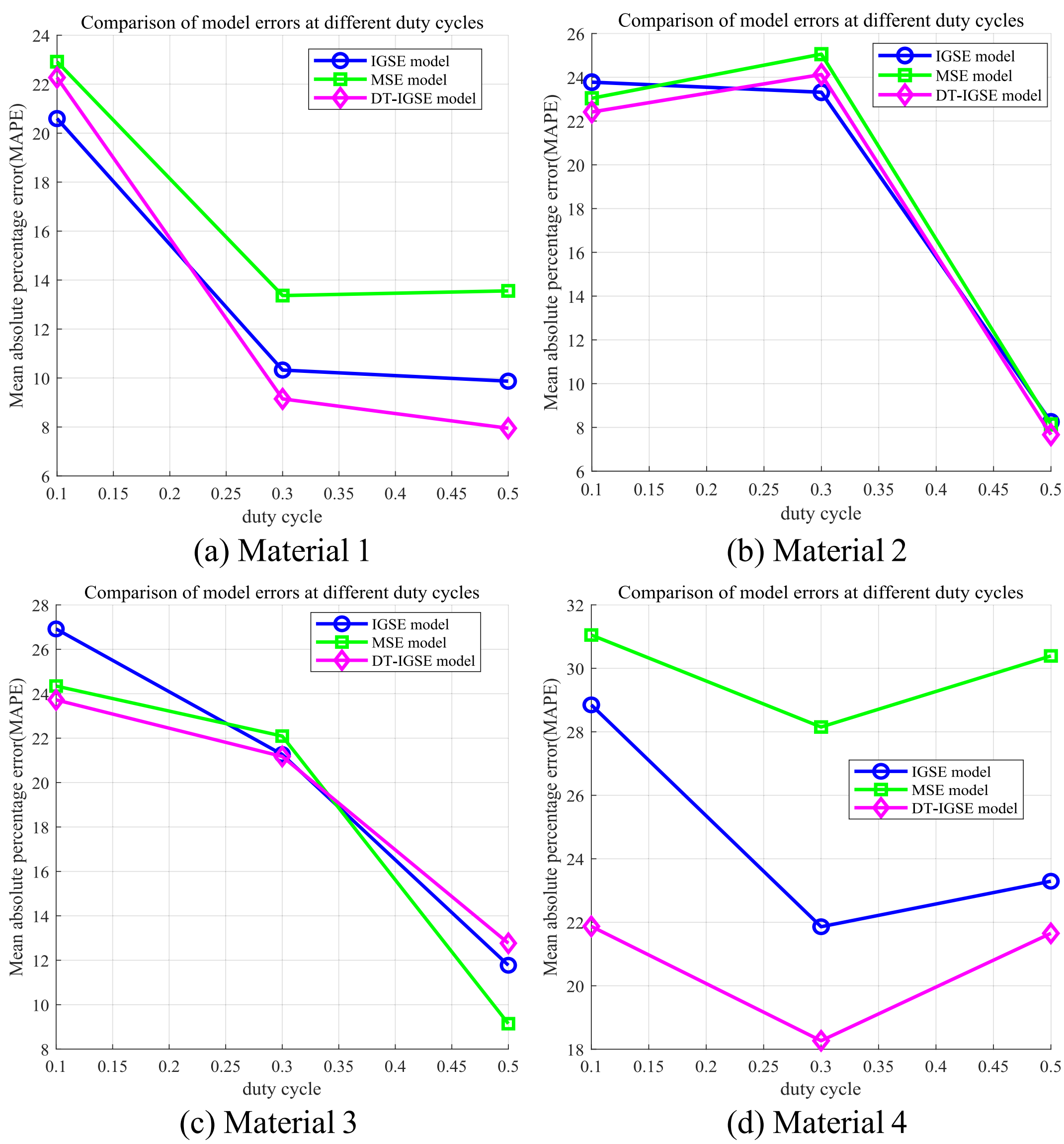

Fig. 4. Line chart of analysis of different duty cycles and model error for four materials.

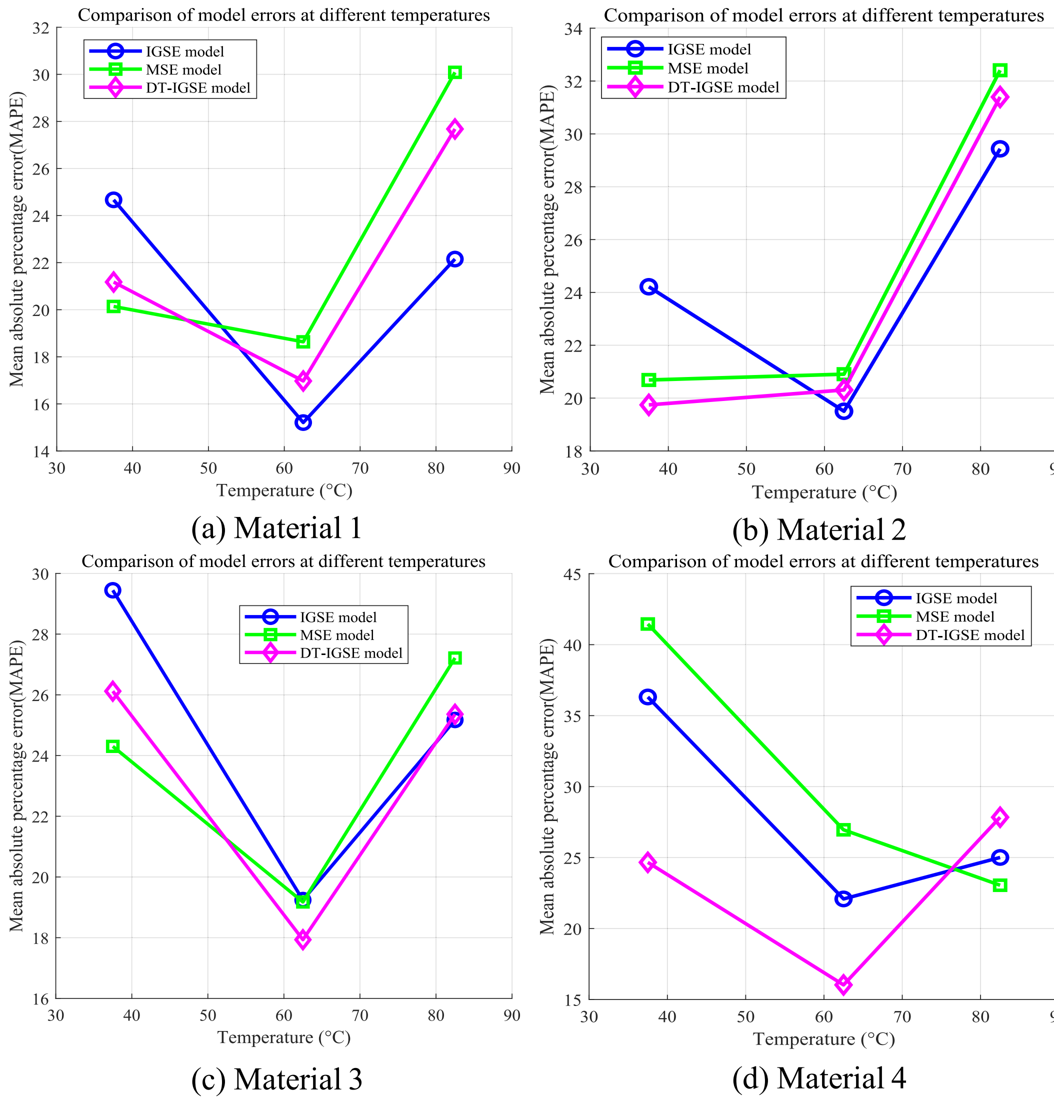

Fig. 5. Line chart of temperature vs. equations error analysis for four materials.

shows an anomaly. This discrepancy may stem from experimental data inaccuracies. However, Material 1 performs well in error analysis across varying duty cycles yet shows mediocre results at different temperatures. This suggests that temperature correction factors, when applied as adjustments to duty cycles or frequencies, may reduce overall error for this material. Consequently, a coupling relationship between temperature and certain material coefficients is suspected. Analyzing this coupling could further refine the relationship between temperature terms and empirical models.

Fig. 6 presents a three-dimensional comparison of prediction model errors, revealing that the combined correction of DT-IGSE significantly reduces the overall prediction error for magnetic core losses. This also offers a novel correction approach for magnetic loss prediction models: when developing correction models based on duty cycle or frequency, it is essential to simultaneously account for temperature's impact on materials and the resulting prediction errors introduced to the overall model.

3. Conclusion

This paper first compares the prediction errors of various models for core losses under triangular excitation waveforms, concluding that the IGSE model is more suitable for predicting core losses under triangular excitation waveforms. Therefore, a new prediction model, DT-IGSE, is proposed based on the IGSE equation. This method first analyzes the waveform composite hypothesis and introduces a duty cycle correction factor to account for minor errors during superposition. Considering that temperature may exhibit coupling relationships with material or frequency variables, it incorporates temperature as a significant factor influencing core loss. After analyzing the three-dimensional surface relationship between duty cycle, temperature, and core loss, a temperature correction factor is ultimately added to the duty cycle correction. Multi-parameter fitting analysis demonstrated that DT-IGSE yields higher agreement between predicted and actual values. Comparative analysis of four material sets also revealed DT-IGSE exhibits robust performance across different materials.

This paper also proposes a novel perspective for refining empirical models of magnetic loss, namely, a joint correction considering the coupled relationship between temperature, frequency, and duty cycle. Since temperature may alter material properties and consequently cause changes in certain independent variables within empirical models, incorporating temperature correction can offset errors arising from such variations. This paper directly incorporates temperature correction terms through three-dimensional image analysis. Future research could classify ferromagnetic materials based on material coefficients to explore additional factors influencing temperature effects under

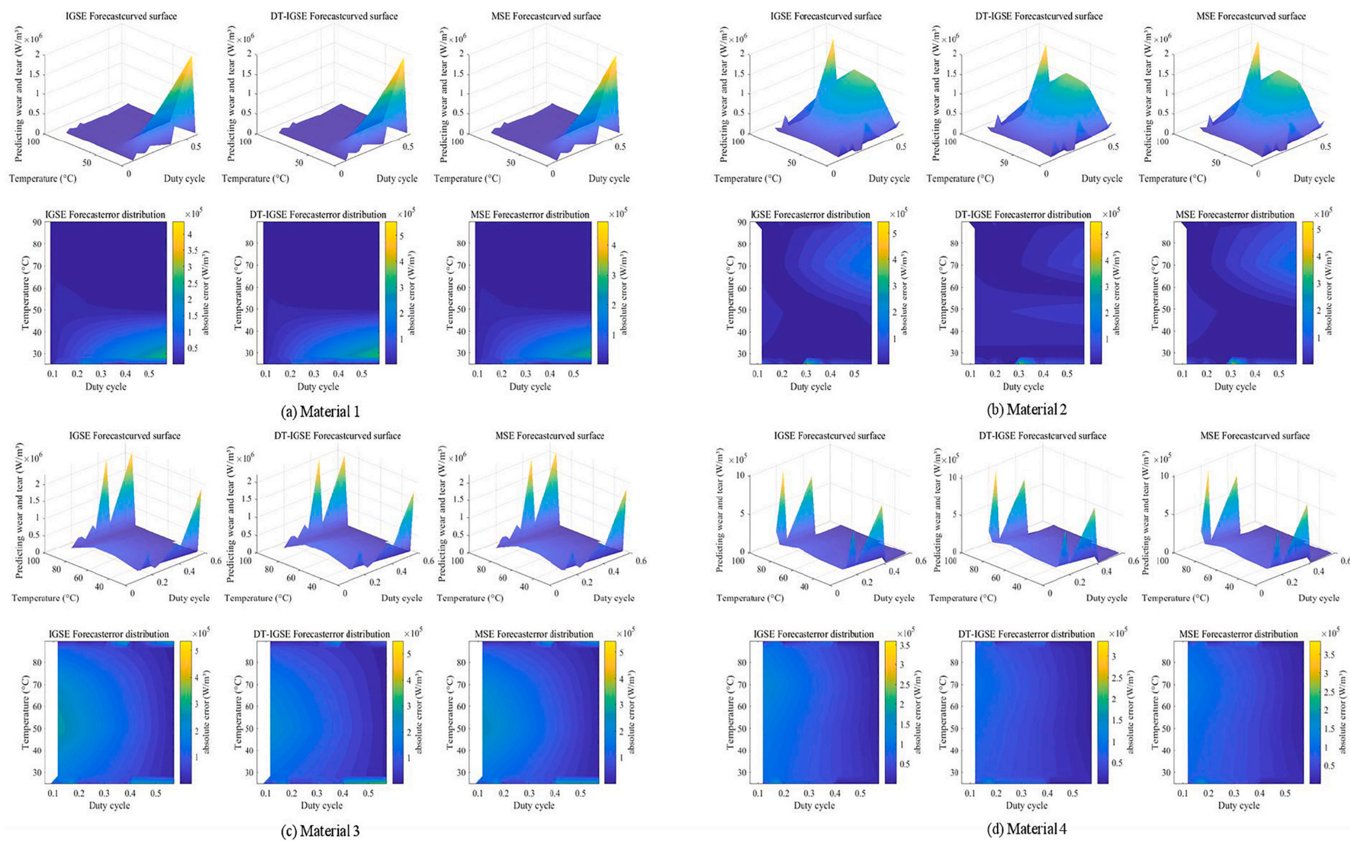

Fig. 6. Comparison of three-dimensional curves for temperature-dependent core losses at different material duty cycles and three prediction equations.

different coefficients. For instance, investigating the angle at which temperature affects ferromagnetic molecular motion rates could yield more precise coupling relationships, thereby further refining model errors.

CRediT authorship contribution statement

Yue Wang: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Xiaodong Liu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis. **Jingwei Li:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] G. Bertotti, General properties of power losses in soft ferromagnetic materials, IEEE Trans. Magn. 24 (1) (1988) 621–630.
[2] C.P. Steinmetz, On the law of hysteresis, Proc. IEEE 72 (2) (1984) 197–221.
[3] R. Severns, HF-core losses for nonsinusoidal waveforms, In *Proc. HFPC* 91 (1991) 140–148.
[4] K. Venkatachalam, C.R. Sullivan, T. Abdallah, H. Tacca, "Accurate prediction of ferrite core loss with nonsinusoidal waveforms using only Steinmetz parameters," 2002 IEEE workshop on computers in power electronics, 2002. Proceedings, Mayaguez, PR, USA (2002) 36–41.
[5] J. Reinert, A. Brockmeyer, R.W.A.A. De Doncker, Calculation of losses in ferro- and ferrimagnetic materials based on the modified Steinmetz equation, IEEE Trans. Ind. Appl. 37 (4) (2001) 1055–1061.
[6] T. Jieli Li, Abdallah, C.R. Sullivan, Improved calculation of core loss with nonsinusoidal waveforms, in: Conference Record of the 2001 IEEE Industry Applications Conference. 36th IAS Annual Meeting (Cat. No.01CH37248), Chicago, IL, USA vol. 4, 2001, pp. 2203–2210.
[7] WILSON, E., Magnetic induction in Iron and other metals, Nature 47 (1893) 460–461.
[8] A. Arruti, J. Anzola, F.J. Pérez-Cebolla, I. Aizpuru, M. Mazuela, The composite improved generalized Steinmetz equation (ciGSE): an accurate model combining the composite waveform hypothesis with classical approaches, IEEE Trans. Power Electron. 39 (1) (2024) 1162–1173.
[9] S. Barg, S. Barg, K. Bertilsson, A review on the empirical Core loss models for symmetric flux waveforms, IEEE Trans. Power Electron. 40 (1) (2025) 1609–1621.
[10] W. Zhang, Q. Yang, Y. Li, Z. Lin, M. Yang, Temperature dependence of powder cores magnetic properties for medium-frequency applications, IEEE Trans. Magn. 58 (2) (2022) 1–5.
[11] S. Liu, T. Li, J. Luo, Y. Yang, C. Liu, Y. Wang, Core loss analysis of soft magnetic composites based on 3D model considering temperature influence, IEEE Access 9 (2021) 153420–153428.
[12] S. Barg, K. Ammous, H. Mejbri, A. Ammous, An improved empirical formulation for magnetic Core losses estimation under nonsinusoidal induction, IEEE Trans. Power Electron. 32 (3) (2017) 2146–2154.
[13] C.R. Sullivan, J.H. Harris, E. Herbert, "Core loss predictions for general PWM waveforms from a simplified set of measured data," 2010 Twenty-fifth annual IEEE applied power electronics conference and exposition (APEC), Palm Springs, CA, USA, 2010, pp. 1048–1055.
[14] H. Li, et al., "MagNet: A Machine Learning Framework for Magnetic Core Loss Modeling," in proc. IEEE Workshop on Control and Modelling of Power Electronics (COMPEL), 2020, pp. 1–8.

[15] D. Serrano, et al., "Neural Network as Datasheet: Modeling B-H Loops of Power Magnetics with Sequence-to-Sequence LSTM Encoder-Decoder Architecture," 2022 IEEE 23rd workshop on control and Modeling for power electronics (COMPEL), Tel Aviv, Israel, 2022, pp. 1–8.
[16] Yongjian Li, Kangshuo Qin, Shenghui Mu, Jiatong Yin, A core loss estimation method based on improved waveform coefficient Steinmetz equation for asymmetric triangular flux density waveform, AIP Adv. 14 (1) (2024) 015121.
[17] S. Yue, Y. Li, Q. Yang, X. Yu, C. Zhang, Comparative analysis of core loss calculation methods for magnetic materials under nonsinusoidal excitations, IEEE Trans. Magn. 54 (11) (2018) 1–5, 6300605.
[18] S. Barg, K. Bertilsson, Core loss Modeling and calculation for trapezoidal magnetic flux density waveform, IEEE Trans Ind Electron 68 (9) (2021) 7975–7984.
[19] H.Y. Lu, J.G. Zhu, S.Y.R. Hui, Measurement and Modeling of thermal effects on magnetic hysteresis of soft ferrites, IEEE Transactions on Magnetics 43 (11) (2007) 3952–3960.
[20] C.P. Steinmetz, On the law of hysteresis, Trans. Am. Inst. Electr. Eng. IX (1) (1892) 1–64.
[21] H. Zhao, H.H. Eldeeb, Y. Zhang, Y. Zhan, G. Xu, O.A. Mohammed, "An Improved Core Loss Model of Ferromagnetic Materials Considering High-Frequency and Non-Sinusoidal Supply," 2020 IEEE industry applications society annual meeting, Detroit, MI, USA, 2020, pp. 1–6.
[22] D.C. Jiles, Frequency dependence of hysteresis curves in conducting magnetic materials, J. Appl. Phys. 76 (1994) 5849–5855.